

From a PV Model to P&O and PSO

A concise visual guide to building and testing an MPPT simulation

The question this guide answers

If we begin with only a few solar-module datasheet values, how do we create a believable power curve and then use that curve to test Perturb and Observe (P&O) and Particle Swarm Optimisation (PSO)?

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Teaching example: Shell SQ150-PC module at standard test conditions

1 The whole idea

An MPPT algorithm makes decisions from measured power. To test those decisions in software, we first need a plant: a mathematical model that tells us what current and power would result from a chosen PV voltage. Without that relationship, the algorithm has nothing meaningful to observe.

Figure 1 shows the complete teaching loop. Datasheet values anchor the PV model. The model produces the current–voltage (I–V) and power–voltage (P–V) curves. P&O or PSO then proposes a voltage reference, receives the resulting voltage and current, calculates power, and makes its next decision.

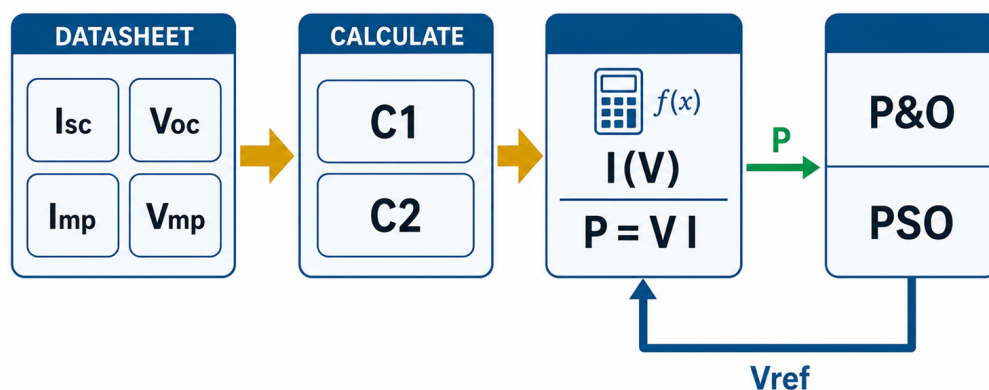


Figure 1: The modelling and MPPT simulation chain. The algorithm receives evaluated power and returns a voltage reference to the empirical plant.

The central idea

The optimiser does not create more solar power. It changes the requested operating point and checks whether the available power increased or decreased.

The simulation repeats four simple actions:

1. apply a candidate PV voltage reference;
2. ask the plant model for current and calculate power;
3. store the result as one settled measurement; and
4. let the MPPT algorithm choose the next candidate, then repeat.

The calculated MPP is shown on plots so that we can judge the algorithm afterward. It is not passed into P&O or PSO. The algorithm receives only its candidate voltage and the corresponding current and power.

Simulation, renderer, and hardware are different. The numerical trace is the simulation result; a table, plot, or GIF only displays it. The scripts assume an ideal inner loop that places the module at the requested voltage. They do not simulate switching, converter transients, a load, sensors, or PWM timing.

2 Build and verify the PV curve

2.1 Start from four datasheet anchors

The Shell SQ150-PC example is defined at standard test conditions (STC): a cell temperature of 25 °C and irradiance of 1000 W/m². Four datasheet quantities describe the ends and the rated peak of the curve.

Table 1: Shell SQ150-PC datasheet anchors used by the empirical model

Quantity	Symbol	Value	Meaning
Short-circuit current	I_{SC}	4.82 A	Current when terminal voltage is zero
Open-circuit voltage	V_{OC}	43.46 V	Voltage when output current is zero
MPP current	I_{MP}	4.48 A	Current at the rated maximum-power point
MPP voltage	V_{MP}	33.73 V	Voltage at the rated maximum-power point

These values are not yet a continuous plant. The MPPT algorithm may request any voltage between zero and V_{OC} , so the model must supply the missing points between the datasheet anchors. The prescribed reading gives a convenient two-coefficient empirical approximation for this purpose.

2.2 Calculate the C1–C2 curve coefficients

The coefficients C_1 and C_2 are dimensionless numbers that set the shape of the curve. They are calculated directly from the four datasheet quantities:

$$C_2 = \frac{V_{MP}/V_{OC} - 1}{\ln(1 - I_{MP}/I_{SC})}, \quad (1)$$

$$C_1 = \left(1 - \frac{I_{MP}}{I_{SC}}\right) \exp\left(-\frac{V_{MP}}{C_2 V_{OC}}\right).$$

For the Shell module, first calculate the fractional current drop: $1 - I_{MP}/I_{SC} = 1 - 4.48/4.82 = 0.0705394$. The voltage ratio is $V_{MP}/V_{OC} = 33.73/43.46 = 0.776116$, and the logarithm of the current drop is -2.651584 . Substitution into Equation 1 gives $C_2 = 0.0844341$ and $C_1 = 7.18469 \times 10^{-6}$.

Why this method is suitable here

It uses only the four usual STC datasheet values. Calculate C_1 and C_2 once, then evaluate current directly at each requested voltage. Every step can be reproduced with a scientific calculator or a short script.

2.3 Evaluate current, then calculate power

For every requested voltage between zero and V_{OC} , use the empirical current equation and multiply voltage by current:

$$I(V) = I_{SC} \left[1 - C_1 \left(\exp \left(\frac{V}{C_2 V_{OC}} \right) - 1 \right) \right],$$

$$P(V) = VI(V),$$

$$V_{MPP} = \arg \max_{0 \leq V \leq V_{OC}} P(V).$$
(2)

The implementation creates 1,601 equally spaced voltage values from zero to 43.46 V. It evaluates $I(V)$ at each value, calculates $P(V)$, and selects the largest sampled power. Each voltage therefore gives one current value immediately.

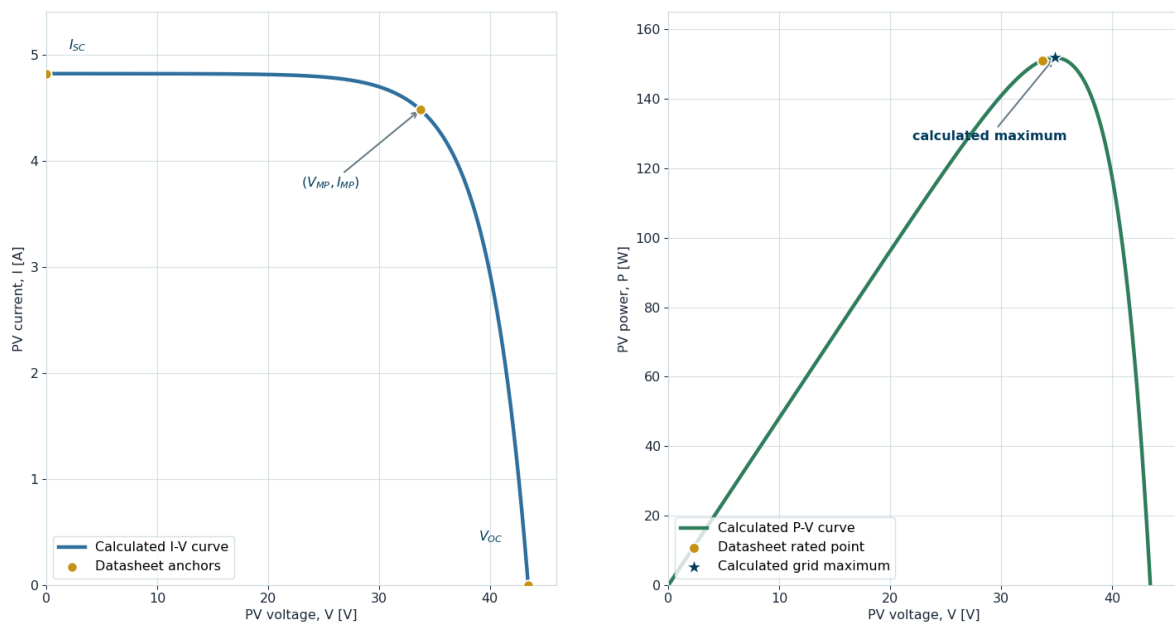


Figure 2: The C1–C2 current equation gives the I–V curve on the left; $P = VI$ gives the P–V curve on the right. The datasheet rated point is shown for comparison with the calculated grid maximum.

The raw equation gives $I(0) = 4.82$ A and $I(33.73 \text{ V}) = 4.4800346$ A. At V_{OC} it leaves only $34.6 \mu\text{A}$, so the plotting code enforces the known open-circuit endpoint as exactly zero. The sampled maximum is 34.8223 V, 4.3621 A, and 151.9001 W.

The empirical peak is 0.7897 W (about 0.52%) above the datasheet rating of 151.1104 W and about 1.09 V higher. That small difference is a limitation of this simple interpolation, not extra power created by the model. Both algorithms receive this same calculated curve.

Checkpoint before adding an MPPT algorithm

Confirm the short-circuit and open-circuit endpoints, check the current at the rated MPP, and compare the calculated maximum with the datasheet rating. The calculated maximum is benchmark information for evaluation; it is not supplied to the MPPT algorithm.

3 Add Perturb and Observe

P&O is a local search. It nudges the operating voltage, observes the resulting power, and uses that change to choose the next direction. It needs little memory: the previous voltage, previous power, present direction, and step size.

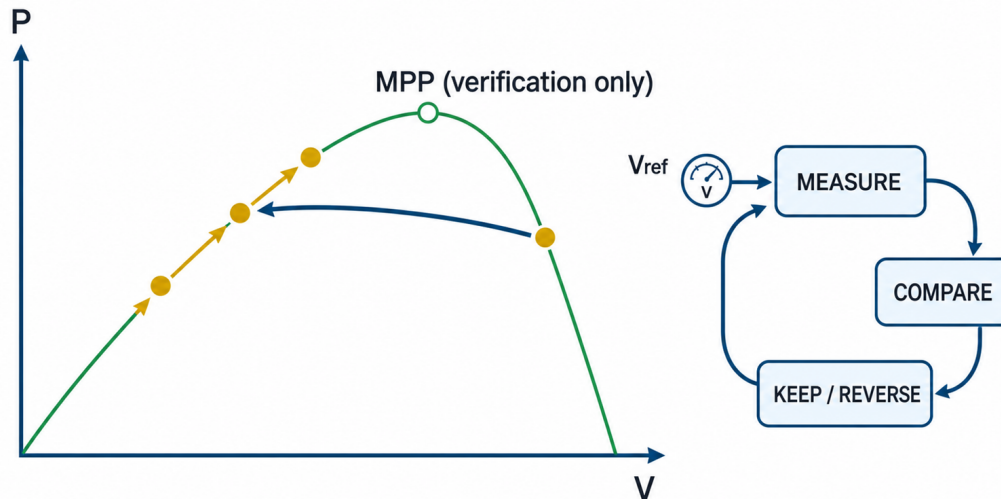


Figure 3: P&O uses consecutive settled measurements. The plotted MPP is a verification reference and is not supplied to the algorithm.

The decision can be written compactly as

$$\begin{aligned} \Delta V &= V[k] - V[k-1], & \Delta P &= P[k] - P[k-1], \\ q_{\text{next}} &= \begin{cases} \text{sign}(\Delta V), & \Delta P \geq 0, \\ -\text{sign}(\Delta V), & \Delta P < 0, \end{cases} \\ V_{\text{ref}}[k+1] &= \text{clip}(V[k] + q_{\text{next}} \Delta V_{\text{step}}). \end{aligned} \quad (3)$$

Table 2: The P&O decision in two rows

Observed change	Interpretation	Next action
$\Delta P \geq 0$	The last move did not reduce power.	Keep the same voltage direction.
$\Delta P < 0$	The last move reduced power.	Reverse the voltage direction.

Listing 1: P&O teaching pseudocode

```

choose Vref, step, and initial direction
measure I and calculate P = Vref * I
repeat:
    apply Vref + direction * step
    measure the new I and P
    if power decreased: reverse direction
    store the new V and P
    clip the next Vref to the allowed range

```

Worked P&O reversal

At 33.0 V the model gives 149.8652 W. A positive probe moves to 35.5 V, where power rises to 151.5592 W. Because $\Delta P = +1.6940$ W, P&O keeps moving upward. The next probe at 38.0 V gives only 141.7969 W, so $\Delta P = -9.7623$ W. P&O now reverses and commands 35.5 V.

With a fixed 2.5 V step, the teaching trace cannot land exactly on 34.8223 V. It moves to either side of the peak and repeatedly corrects itself. This expected hunting illustrates the trade-off: a larger step moves quickly but produces a wider steady oscillation; a smaller step is gentler but takes more measurements to cover the same distance.

Teaching P&O versus the supplied firmware

This Python example directly requests PV voltage and assumes ideal settling. The STM32 implementation perturbs converter gain, waits for the new operating point, uses an eight-sample trimmed mean, rejects small changes with a deadband, adjusts its step size, and finally maps gain to buck and boost duty commands. The example teaches the core decision; it does not reproduce the firmware timing or power stage.

4 Add Particle Swarm Optimisation

PSO searches with several candidate solutions instead of one local probe. In this one-dimensional example, each particle position is a candidate PV voltage. Its fitness is the power measured at that voltage.

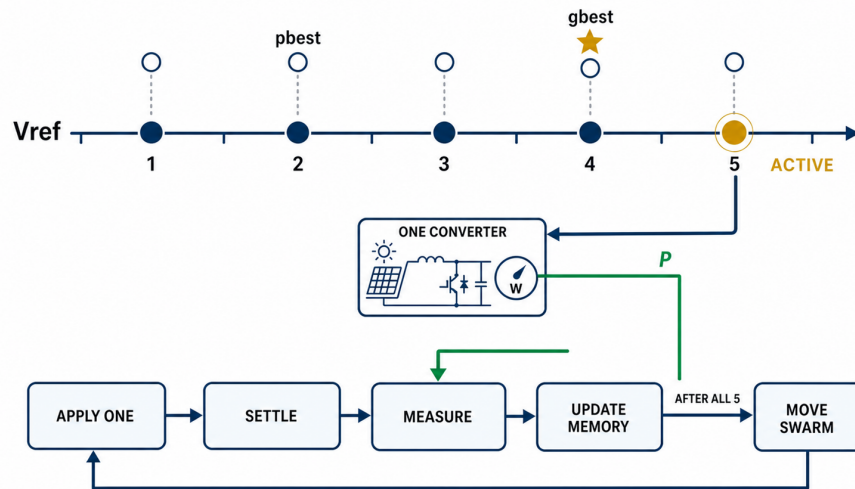


Figure 4: PSO evaluates candidate voltages sequentially, remembers personal and global bests, and then moves the swarm.

Table 3: Minimum PSO vocabulary

Term	Meaning in this simulation
Position x_i	Particle i 's candidate voltage reference
Velocity v_i	Its proposed voltage movement per generation
Fitness	The power $P(x_i)$ obtained at the candidate voltage
Personal best	The best position particle i has measured so far
Global best	The best position measured by any particle so far
Generation	One complete candidate scan followed by one swarm movement

After a generation has been measured, particle i is moved by

$$v'_i = wv_i + c_p r_p (pbest_i - x_i) + c_g r_g (gbest - x_i), \quad x'_i = \text{clip}(x_i + \text{clip}(v'_i)). \quad (4)$$

The three velocity terms have clear roles. Inertia wv_i carries some previous motion forward. The personal term draws the particle toward its own best result. The global term draws it toward the best result shared by the swarm. Random factors r_p and r_g vary the strength of the two attractions.

Listing 2: PSO teaching pseudocode

```

randomly initialise five positions and velocities
repeat for three generations:
  for each particle, one at a time:
    apply Vref, settle, and measure power
    update its pbest and the shared gbest
  if another generation remains:
    update and clip velocities, then move all particles
apply and measure the final gbest once

```

Worked movement for particle 5

After generation 1, particle 5 is at 29.7040 V with velocity +1.8450 V/generation. Its personal best is the same position, while the best measured swarm position is 31.1413 V. Its inertia, personal, and global contributions are +1.1993, 0, and +0.3457 V/generation. The new velocity is therefore +1.5449, which moves the particle to 31.2490 V. Its next 145.2175 W measurement improves its personal best, but not the global best.

The best measured result progresses from 144.8729 W at 31.1413 V in generation 1, to 150.5034 W at 33.3378 V in generation 2, and to 151.8968 W at 34.7655 V in generation 3.

The supplied trace uses seed 40, five particles, three generations, $w = 0.65$, $c_p = c_g = 1.4$, a search range of 4.5–41.0 V, and a 6.0 V/generation velocity limit. It makes 15 sequential candidate measurements, then applies and measures the final global best once: 16 in total.

What the PSO animation represents

The particle positions are updated together in the mathematical generation step, but a single physical converter cannot occupy five operating points at once. Each candidate is applied, allowed to settle, and measured in sequence. The swarm moves only after all five candidates have been tested; the final global best is applied once after generation 3. These deterministic settings were chosen for a clear teaching story. This is conventional PSO, not the PSO–Quasi-Newton method from the prescribed paper.

5 Run, verify, and interpret

Install the small Python dependency set once, then run the three numerical checks from the repository root:

Listing 3: Commands used to reproduce and check the teaching model

```
python -m pip install -r requirements-pv-animation.txt
python tools/make_pv_empirical_curves.py --validate-only
python tools/make_pv_po_animation.py --validate-only
python tools/make_pv_pso_animation.py --validate-only
```

The validators stop with an error if a documented result changes. A successful run should pass these four checkpoints.

Table 4: Numerical checkpoints for the supplied simulations

Checkpoint	Expected result
Datasheet endpoints	$I(0) = 4.82 \text{ A}$ and the enforced endpoint is $I(V_{OC}) = 0$
Calculated grid MPP	34.8223 V, 4.3621 A, 151.9001 W
P&O reversal	The 38.0 V probe loses power, so the next command is 35.5 V
PSO improvement	The generation best rises from 144.8729 to 150.5034 to 151.8968 W

5.1 Interpret what the traces show

Table 5: What the two teaching traces demonstrate

Feature	P&O	PSO
Search style	One local voltage step at a time	Several candidate voltages per generation
Memory	Previous measurement and direction	Each personal best plus one global best
This trace	11 measurements	16 measurements
Main limitation	Fixed-step hunting near the peak	More measurements and chosen tuning constants

Included and not included

Included: one Shell SQ150-PC module at fixed STC, one smooth peak, the C1–C2 empirical curve, and ideal voltage-reference control.

Not included: converter and load dynamics, switching, settling error, sensor noise, changing irradiance or temperature, and partial shading. Those effects must be added before drawing conclusions about hardware performance.

For two-diode modelling, converter duty-cycle optimisation, environmental studies, and PSO–Quasi-Newton extensions, continue with the supplied [ENR 420 Research Module — Energy Systems](#). This companion guide deliberately stops at the simpler datasheet-to-curve-to-algorithm workflow.